COMP 110 Midterm Study Guide

Chapters 1 to 5

Name _____ Date _____

Use this guide to review concepts, trace Java code, repair errors, and design short programs. Work by hand first, then compile and run a test version. Show intermediate values and explain your reasoning. Use separate paper for longer programs.

Scope and assumptions. This is practice material and does not specify what will or will not appear on the exam. Unless a question says otherwise, use Java 17, assume valid token input, and interpret the shown snippets inside main with needed imports. ASCII exercises use ordinary Latin characters and digits. Explain any extra assumptions in your answer.

How to study. For a trace, record each changed variable and exact output. For a program, identify input, processing, output, and tests. For a loop, state initialization, condition, body, progress, and stopping state. Revisit questions you can solve only after running code.

Chapter 1 Computers Programs and Java

Review the IPO model, hardware and software, algorithms, translation, the JVM, program structure, comments, and error categories.

1. Explain the input process output model. For a program that calculates a circle area, identify the input, processing step, and output. State the units.

Response _____
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2. Classify CPU, memory, operating system, compiler, keyboard, and Java application as hardware or software. Explain how the operating system helps applications use hardware.

Response _____
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3. Compare machine language, assembly language, and a high-level language. What is the role of a translator? What does portable Java bytecode require in order to run?

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4. Write a finite, ordered algorithm that reads three scores and reports their average. Include a known-result test before writing Java.

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5. Describe the roles of the JDK, javac, the JVM, source code, and bytecode. Put editing, compiling, and running in order. Give commands to compile and run Welcome.java.

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6. Classify each problem as a compile-time error, a run-time error, or a logic error: a missing semicolon; integer division by zero during a run; a circle-area program using the circumference formula. Explain how each is discovered.

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7. Write a complete Java application named Welcome that prints Welcome to COMP 110 on one line. Include the class declaration, main header, braces, and statement.

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8. This program is intended to compile from Welcome.java. Identify and repair its errors. Explain why a compiler can detect these errors without running the program.

```
public class welcome {
    public static main(String[] args) {
        System.out.println("Hello")
    }
}
```

Response _____

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9. Explain the difference between syntax and meaning. Give one example of valid Java that calculates an unintended result. Explain why successful compilation does not prove correctness.

Response _____

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10. Choose clear names for a circle radius, its area, and an hourly pay rate. Show one useful comment and explain how indentation helps a reader understand blocks.

Response _____

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Chapter 2 Elementary Programming

Review identifiers, primitive types, assignment, arithmetic, integer division, casts, constants, input, numeric range, and formatting.

11. Under Java 17, which names are valid identifiers: total, 2total, hourlyRate, class, _value, and _? For valid names, distinguish legal syntax from a recommended naming convention.

Response _____
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12. Choose suitable types for a student count, a radius with a fractional part, a population above three billion, one ASCII letter, and a true-or-false flag. State any range assumptions.

Response _____
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13. Explain why this code does not compile. Give two different repairs that ensure a value is assigned before it is read.

```
int count;
System.out.println(count);
```

Response _____
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14. Find x and y after this assignment sequence. Explain the order in which values are stored.

```
int x = 1;
int y = x = x + 1;
System.out.println(x + " " + y);
```

Response _____

15. Evaluate each expression and state its result type: 7 / 2; 7 / 2.0; 17 % 5; (double) (7 / 2); (double) 7 / 2; (int) -3.9. Explain why the two cast placements differ.

Response _____
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16. Trace the compound assignments. Is the result the same as adding 9.0 and casting once? Explain.

```
int sum = 0;
sum += 4.5;
sum += 4.5;
System.out.println(sum);
```

Response _____
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17. Predict all output lines. Explain numeric addition, precedence, and text concatenation.

```
System.out.println(2 + 3 * 4);
System.out.println((2 + 3) * 4);
System.out.println("Sum: " + 2 + 3);
System.out.println("Sum: " + (2 + 3));
```

Response _____
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18. Write a program that reads a finite Fahrenheit value and displays Celsius with two decimal places. Use (5.0 / 9) * (fahrenheit - 32). Explain what changes if 5 / 9 is used. Test 32, 212, and -40.

Response _____

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19. Write an input process output plan and a Java calculation for straight-time gross pay. Read finite nonnegative hours and hourly rate. For 8 hours at 18.50, state the expected result. Overtime is outside this question.

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20. Declare a named constant for a hypothetical 6 percent tax rate. Read a nonnegative purchase amount and display tax and total with two decimal places. Explain whether printf changes the stored double value.

Response _____

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21. Predict all three outputs. Explain when widening occurs and which calculation overflows.

```
int largest = 2_147_483_647;
int a = largest + 1;
long b = largest + 1;
long c = (long) largest + 1;
System.out.println(a);
System.out.println(b);
System.out.println(c);
```

Response _____

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22. A three-score program computes double average = total / 3. Choose a valid test that exposes the lost fraction, repair the expression, and include tests at scores 0 and 100. Assume each score is an int from 0 to 100.

Response _____

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Chapter 3 Selections

Review booleans, ranges, branch order, nesting, switch, random ranges, short circuit evaluation, and boundary tests.

23. Evaluate and explain: true || false; true ^ true; !(false && true). State which boolean operators can skip evaluation of their right operand.

Response _____
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24. Write a boolean expression that is true when weight is at least 100 and less than 200. Why is 100 <= weight < 200 invalid Java? Test 99, 100, 199, and 200.

Response _____
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25. Write an if statement that decreases pay by 3 percent when score is less than 90. Assume finite doubles. Explain why multiplying pay by 0.97 differs from subtracting 0.03.

Response _____
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26. A grade classifier checks score >= 60 before score >= 90 in an else-if chain. Explain the result for 95 and repair the order. Give tests immediately below, at, and above the threshold 80.

Response _____
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27. Predict the output and identify which if owns the else. The second message describes the path incorrectly. Rewrite it with braces and a message that accurately describes the selected path.

```
int x = 1, y = -1, z = 1;
if (x > 0)
    if (y > 0)
        System.out.println("x > 0 and y > 0");
    else if (z > 0)
        System.out.println("x < 0 and z > 0");
```

Response _____
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28. What is y after this classic switch? Trace every executed assignment. Then add breaks to make case 0 finish with y equal to 0.

```
int x = 0, y = 0;
switch (x) {
    case 0: y = 0;
    case 1: y = 1;
    default: y = -1;
}
System.out.println(y);
```

Response _____
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29. For day equal to 2, compare the behavior of colon case labels without breaks and arrow switch rules. Write one example of each. Which form can fall through? State the Java version requirement for the arrow form.

Response _____
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30. Write the Gregorian leap-year condition for a positive int year. State the expected results for 2023, 2024, 1900, and 2000. Explain why year % 4 == 0 alone is insufficient.

Response _____
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31. Write one Java expression using Math.random that generates an int from 50 through 100 inclusive. State how many outcomes are possible and justify both endpoints. Explain why casting before scaling is wrong.

Response _____
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32. Predict the output. Why is the operand order important? Explain the effect of replacing && with boolean &.

```
int number = 10, divisor = 0;
boolean result = divisor != 0 && number / divisor > 2;
System.out.println(result);
```

Response _____
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33. Identify the problem in if (flag = true) when flag is boolean. Contrast it with if (value = 0) when value is int. Show a clear boolean test and explain the role of braces and stray semicolons.

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34. Write a selection that classifies three int side lengths as equilateral, isosceles, scalene, or not a triangle. Assume each side is 1 through 1,000,000. Test 3 3 3, 3 3 4, 3 4 5, and 1 2 3. Explain branch order.

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Chapter 4 Math Characters and Strings

Review Math calls, bases, char codes, Unicode limits, printf, String indexes, comparisons, input, parsing, and search guards.

35. Predict the outputs and return types of Math.ceil(-2.3), Math.floor(-2.3), Math.rint(2.5), Math.rint(3.5), Math.round(-2.5), and Math.sqrt(9). Distinguish tie-to-even from ties toward positive infinity.

Response _____
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36. Write a Java calculation for the sine of 30 degrees. Explain why Math.sin(30) does not mean sine of 30 degrees. State how you would display the result to three decimal places.

Response _____
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37. Convert decimal 123 to binary, octal, and hexadecimal. Show quotient and remainder steps for one conversion. Convert binary 10101011 and hex 7F back to decimal using place values.

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38. Explain the difference between the Java literal 010, the string "010", Integer.parseInt("010"), and Integer.parseInt("10", 8). State the numeric results of the executable expressions.

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39. Find the results of (int) '7', '7' - '0', 'A' + 1, and (char) ('A' + 1). Explain why a char is not always a whole visible Unicode character.

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40. Choose Character methods to test for a digit, a letter, or a letter or digit, and to uppercase a char. Explain why Character.isDigit can accept a value outside the ASCII range '0' through '9'.

Response _____
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41. Predict the exact output, using the bars to show padding. Explain minimum width, left alignment, and precision.

```
System.out.printf(java.util.Locale.US, "|%5d|%n", 42);
System.out.printf(java.util.Locale.US, "|%-5d|%n", 42);
System.out.printf(java.util.Locale.US, "%.2f%n", 3.5);
System.out.printf(java.util.Locale.US, "|%3d|%n", 12345);
```

Response _____
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42. For String text = "Java", give length(), charAt(0), charAt(3), substring(1, 3), and substring(4). Which of charAt(4) and substring(4) is valid? Explain the end-exclusive substring rule.

Response _____
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43. Predict the output. Repair the program if its purpose is to print uppercase text.

```
String text = "Java";
text.toUpperCase(java.util.Locale.ROOT);
System.out.println(text);
```

Response _____
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44. A program reads age with nextInt, then a full name with nextLine. Age is on its own line. Explain the pending-line problem and show the appropriate repair. Which method reads one whitespace-delimited token?

Response _____
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45. Predict all results. Explain which test should be used for String contents and how compareTo results should be interpreted.

```
String first = "Java";
String second = new String("Java");
System.out.println(first == second);
System.out.println(first.equals(second));
System.out.println(first.equalsIgnoreCase("java"));
System.out.println("Ada".compareTo("Bob") < 0);
```

Response _____
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46. For "banana", evaluate indexOf('a', 1), indexOf('a', 2), lastIndexOf('a', 3), and indexOf('x'). Explain the inclusive starting index and the meaning of -1. Give a guard before using a search result in substring.

Response _____
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47. Write a program that reads two full city names and prints them in case-sensitive Java String order. Use nextLine and compareTo. Include an equal-name test and a multiword city.

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48. Write a program that accepts exactly one ASCII hexadecimal digit and displays its value 0 through 15. Accept 0-9, A-F, and a-f. Reject empty input, FF, and G before an invalid charAt access. Test every range endpoint.

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Chapter 5 Loops

Review loop forms, counts, progress, sentinels, accumulators, nesting, early exits, numeric algorithms, and text traversal.

49. Trace the loop and give the exact sequence of k values printed. Show j, n % j, and k for each body execution.

```
int j = 1, k = 0, n = 5;
while (j <= n) {
    k = j + (n % j);
    System.out.print(k + " ");
    j++;
}
```

Response _____
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50. Give the number of body executions in each case. Assume n is an int from 1 through 1,000 and the body changes only i as shown. Distinguish body executions from condition checks.

Case	Loop
a	int i = 1; while (i <= n) { i++; }
b	int i = 1; while (i <= 0) { i++; }
c	int i = n; while (i > 0) { i--; }
d	int i = n; while (i >= 0) { i--; }

Response _____
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51. The intended countdown starts with int i = n for n from 1 through 1,000, but uses while (i > 0) { i++; }. Explain the wrong progress direction and repair it. Under actual Java int arithmetic, does overflow eventually make i nonpositive? Do not run the long erroneous loop.

Response _____
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52. Convert this while loop to a for loop and state the final sum. Explain where initialization and update move. Discuss the scope of i if declared in a for header.

```
int sum = 0;
int i = 0;
while (i <= 100) {
    sum += i;
    i = i + 1;
}
```

Response _____
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53. Compare while and do-while when count starts at 5 and the test is count < 3. Give the body-execution count for each. Explain which semicolon is required after a do-while and which semicolon would accidentally empty a while or for body.

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54. Write a loop that reads scores 0-100 until -1, then prints the number of scores and their average. Exclude the sentinel. Handle an immediate -1 and a score of zero. Assume at most 1,000,000 scores followed by -1.

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55. Use nested loops to print five rows of stars, with one star in row 1 through five stars in row 5. State the total star count and where the newline belongs. Explain how an inner loop counter resets.

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56. Write an algorithm and a Java loop that converts pounds to kilograms for pounds 20, 25, 30, ..., 510, 515. Use kilograms = pounds / 2.2 and two decimal places. Show the first two and last two output rows; do not write out all 100 rows in the guide.

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57. Use a while loop to find the smallest nonnegative int n such that n * n is greater than 12,000. State the result and explain why n - 1 fails the condition. Values remain below 1,000 in this exercise.

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58. Read a nonnegative int and calculate its numeric reverse. Keep the original input. Use long for the reverse so every nonnegative int input is supported. Show expected results for 1024, 120, and 0.

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59. Write a loop that computes the decimal digit sum of a nonnegative int. Trace 1024 and explain the roles of % 10 and / 10. State the expected result for input zero.

Response _____

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60. Predict the sum. Explain how continue and break affect the body, the for update, and subsequent iterations.

```
int sum = 0;
for (int i = 1; i <= 6; i++) {
    if (i == 2) continue;
    if (i == 5) break;
    sum += i;
}
System.out.println(sum);
```

Response _____
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61. Outline a program that prints the first 50 primes, ten per line. Distinguish the candidate number, the divisor, and the number of primes printed. Explain why a composite candidate must not increase the prime count.

Response _____
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62. Find the GCD of 16 and 24 with a loop. Describe a brute-force divisor test or Euclidean remainder method, state a useful loop invariant, and include equal-input and coprime tests.

Response _____
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63. Use a loop to compute 1/3 + 3/5 + 5/7 + ... + 95/97 + 97/99. State the number of terms and display six decimal places. Explain why integer division would fail.

Response _____
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64. Write or outline a palindrome test using left and right indexes. Use exact case-sensitive ASCII text, with spaces and punctuation included. Explain the stopping condition and test empty input, one character, noon, Noon, and node.

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65. Integrated practice: read int values until sentinel 0. Report positive count, negative count, total, and average of the nonzero values. Use long for the total and handle no data. Assume at most 1,000,000 int inputs before the sentinel. Give one test for each branch.

Response _____
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Final self check

Can you justify a result without running code, repair the cause of an error, choose a meaningful boundary test, and explain why a loop stops? Use the chapter examples to check your work, then retry a similar question with different values.